1. **Analyze the User's Request:**
    - Previous turn: The user asked for examples of face, degeneracy, and connection maps for $n = 1, 2, 3$ based on a provided image/text. I provided a textual list of tuple mappings.
    - Current turn: The user asks "what are the pictures look like".
    - Intent: The user wants a visual/graphical representation of the mathematical structures described. Specifically, they want to know what the posets $[1]^n$ look like as Hasse diagrams or geometric shapes, and how the maps ∂, σ, γ act on them visually.
    - Constraint: I cannot generate images directly. I must describe them using text (ASCII art, mathematical diagrams, and textual descriptions of geometric shapes).
2. **Deconstruct the Mathematical Objects:**
    - **Object:** $[1]^n$. This is the n-fold product of the poset $[1] = \{0 \leq 1\}$.
    - **Geometric shape:** It represents the n-dimensional hypercube (also called n-cube) including the interior. Its elements are the vertices (corners) of the hypercube, represented by tuples of 0 and 1.
    -
        - $n = 0$: A single point.
    -
        - $n = 1$: A line segment (2 vertices: 0, 1).
    -
        - $n = 2$: A square (4 vertices: $(0, 0), (1, 0), (0, 1), (1, 1)$).
    -
        - $n = 3$: A cube (8 vertices).
    - **Maps and their visual interpretation:**
        - **Faces (∂):** Maps $n - 1$ dimensional cube to n-dimensional cube, embedding it as a boundary. In a square, $\partial_{1,0}$ puts the 1D segment on the left side ($x = 0$), $\partial_{2,1}$ puts it on the top ($y = 1$), etc.
        - **Degeneracies (σ):** Maps n-cube to $n - 1$-cube by collapsing one dimension. For a cube, this is like projecting the square down to a line segment. The formula removes one coordinate.
        - **Connections (γ):** Maps n-cube to $n - 1$-cube by identifying (folding) two adjacent faces. The formula applies $\max$ to two adjacent coordinates. This collapses a square into a line segment in a different way than degeneracy. Standard σ collapses a whole dimension (e.g., x variable disappears), while γ folds along the diagonal of a face.
3. **Formulate the Response Strategy:**
    - Acknowledge that I cannot generate actual pictures, but I can provide detailed descriptions and ASCII/Unicode diagrams.
    - Describe the *shapes* for $n = 1, 2, 3$ (Line, Square, Cube) using the standard hypercube visualization.
    - Provide Hasse diagrams (poset diagrams) for $[1]^n$ for $n = 1, 2, 3$. This shows the order relations (monotonicity) between tuples (e.g., $(0, 0) \leq (1, 0) \leq (1, 1)$).
    - Describe visually where the face maps, degeneracies, and connections go.
        - For example, σ_1^2 (removing x_1) collapses a square into a line by projecting onto the y-axis.
        - γ_1^2 ($\max\{x_1, x_2\}$) collapses a square into a line by folding it along the diagonal from $(0, 1)$ to $(1, 0)$.
    - Use monospaced text format to create diagrammatic representations.
4. **Drafting the Content (Mental Sandbox):**
    - *Hasse diagram of [1]^2 (Square):*

```
(1,1)
 /\
/  \
```

(1,0) (0,1)
\ /
/
(0,0)

```
Wait, strictly speaking in product poset: \((0,0) \le (1,0)\) and \((0,0) \le (0,1)\), and \((1,0) \le
(1,1)\), and \((0,1) \le (1,1)\). So it's a diamond shape.
*   *Hasse diagram of [1]^3 (Cube):*
(0,0,0) bottom. Goes up to (1,1,1) top. The next level has 3 nodes: (1,0,0), (0,1,0), (0,0,1). The level
below top has 3 nodes: (1,1,0), (1,0,1), (0,1,1). It's a 3D diamond (a cube drawn schematically).
*   *Visualizing maps:*
*   For a square (n=2):
    *   \(\partial_{1,0}\) maps line to left edge. \(\partial_{2,1}\) maps line to top edge.
    *   \(\sigma_1\) collapses the square horizontally (onto y-axis). \(\sigma_2\) collapses vertically
(onto x-axis).
    *   \(\gamma_1\) collapses the square onto a line segment by folding along the anti-diagonal (from (0,1)
to (1,0)). Can describe it precisely.
```

5. **Refine the Output:**
   - Structure clearly: n=1, n=2, n=3.
   - Use text-based diagrams (ASCII art). The square and cube are easy to draw schematically.
   - Match the definitions to the visual.
6. **Writing the specific ASCII diagrams:**
   - **n=1**:
     `0 <---> 1` (Just a line segment/arrow).
   - **n=2 (Square)**:

     ```
     (1,0) -------- (1,1)
       |              |
       |              |
     (0,0) -------- (0,1)
     ```

     (Wait, the tuple notation: x_1 is first, x_2 is second. So $(1,0)$ is right-bottom. $(0,1)$ is left-top. Let's draw it properly).
     Let's draw a standard coordinate plane.

     ```
             y
             ^
       (0,1) --- (1,1)    (Top edge)
         |         |
     ```

```
    |         |
  (0,0) --- (1,0)  > x  (Bottom edge)
```

- **n=3 (Cube)**:
  Draw a cube wireframe. It's hard to do perfectly in ASCII, but I can do a standard projection. Or just describe the 3 levels of the Hasse diagram.
  Level 0: (0,0,0)
  Level 1: (1,0,0), (0,1,0), (0,0,1)
  Level 2: (1,1,0), (1,0,1), (0,1,1)
  Level 3: (1,1,1)
  - **Let's try a projected ASCII cube:**

```
      (1,0,1)----------(1,1,1)
     /                 /|
    /                 / |
  (0,0,1)----------(0,1,1)|
   |                |   |
   |    (1,0,0)------|--(1,1,0)
   |    /           | /
   |   /            |/
  (0,0,0)----------(0,1,0)
```

  This is a good representation. Note that the coordinates aren't perfectly aligned to the edges, but it represents the combinatorial structure.

7. **Explain the Maps Visually:**
   - **Faces (∂):** Embedding the lower dimension into the boundary.
     - In Square: $\partial_{1,0}$ is the left vertical edge (x=0), $\partial_{1,1}$ is the right vertical edge (x=1), $\partial_{2,0}$ is the bottom horizontal edge (y=0), $\partial_{2,1}$ is the top horizontal edge (y=1).
   - **Degeneracies (σ):** Projection. Pressing a 2D square flat onto a 1D line.
     - σ_1^2 projects a square onto the y-axis (drop x_1). So $(1,0)$ and $(1,1)$ both map to (1). It's a vertical shadow.
     - σ_2^2 projects onto the x-axis (drop x_2).
   - **Connections (γ):** Folding.
     - For a square, $\gamma_1^2(x_1, x_2) = (\max(x_1, x_2))$. This folds the square into a line segment.
     - Visual: Points $(1,0)$ and $(0,1)$ both map to (1). The square is folded along the diagonal line connecting $(0,1)$ and $(1,0)$. The diagonal line becomes the point (1) on the target line, while $(0,0)$ maps to (0) and $(1,1)$ maps to (1).
8. **Refine Tone and Clarity:**
   - Make it clear I can't draw, but here's exactly how to draw it.
   - Use bullet points and bold text for emphasis.
   - Ensure the Hasse diagrams use standard poset conventions (lines denote $\leq$ relations).
9. **Self-Correction/Verification on the "Picture" concept:**
   - The user asked "what are the pictures look like" - literally. I'll start by clarifying the geometric interpretation of $[1]^n$.

- $[1]^1$ is a line segment.
- $[1]^2$ is a square.
- $[1]^3$ is a cube.
- The maps act on the *entire* cubical complex (vertices and interiors, though since it's a poset of tuples, the "interior" isn't visual unless we consider geometric realization).
- Wait, the text says "objects of □ are posets of the form $[1]^n$". Yes, this is the cubical category (cubes). So $[1]^n$ is the *geometric n-cube*. The faces, degeneracies, and connections are the structure maps of a cubical set.
- Let's emphasize the geometric realization: A line, a square, a cube.

10. **Final Polish:** Provide clear ASCII art for the 3 shapes (n=1, 2, 3) and precise explanations of what the maps look like on them. Include the Hasse diagram for $n = 3$ as well, or at least the coordinate mappings.

    Let's write the final response now.

I cannot create or generate actual picture files, but I can describe exactly what these objects look like geometrically and provide precise ASCII/Unicode diagrams so you can visualize them.

These objects $[1]^n$ are the **geometric n-cubes** (including their vertices). The "face," "degeneracy," and "connection" maps are the structural operations used to build, collapse, or fold these cubes.

Here is what the pictures look like for $n = 1, 2,$ and 3:

$n = 1$: The Line Segment

This is a 1-dimensional cube (an edge). It has two vertices: 0 and 1.

- **Diagram:** `0 — 1`
- **Faces ($\partial_{1,0}, \partial_{1,1}$)**: Map a point ($n = 0$) to the left vertex (0) or the right vertex (1).

$n = 2$: The Square

This is a 2-dimensional cube. It has four vertices (x_1, x_2): $(0, 0), (1, 0), (0, 1), (1, 1)$.

- **Visual (Coordinate Grid):**

```
        y-axis
          ^
(0,1) --------- (1,1)
  |               |
  |    SQUARE     |
  |               |
(0,0) --------- (1,0)  > x-axis
```

- **Faces (∂)**: These map a 1D segment to one of the 4 boundary edges of the square.
  - $\partial_{1,0}$: Left edge (where $x_1 = 0$)

    - $\partial_{1,1}$: Right edge (where $x_1 = 1$)
    - $\partial_{2,0}$: Bottom edge (where $x_2 = 0$)
    - $\partial_{2,1}$: Top edge (where $x_2 = 1$)
- **Degeneracies (σ)**: These *collapse* the 2D square down to a 1D line segment (like casting a shadow).
    - σ_1^2: Projects the square straight down onto the y-axis (dropping x_1).
    - σ_2^2: Projects the square straight sideways onto the x-axis (dropping x_2).
- **Connections (γ)**: This folds the square into a 1D line segment, but differently than a projection.
    - γ_1^2: This folds the square **along the diagonal** connecting $(1,0)$ and $(0,1)$.
    - The point $(0,0)$ maps to 0, $(1,1)$ maps to 1, and the whole diagonal line $(1,0) \leftrightarrow (0,1)$ folds down to become a single point (1).

$n = 3$: The Cube

This is a 3-dimensional cube. It has 8 vertices represented by tuples (x_1, x_2, x_3).

- **Visual (3D Projection / Wireframe):**

```
      (1, 0, 1)----------(1, 1, 1)
     /                   /|
    /                   / |
(0, 0, 1)----------(0, 1, 1) |
   |                   |  |
   |  (1, 0, 0)--------|--(1, 1, 0)
   | /                 | /
   |/                  |/
(0, 0, 0)----------(0, 1, 0)
```

- **Faces (∂)**: These embed a 2D square onto the boundaries of the 3D cube. There are 6 faces in total. For example:
    - $\partial_{1,0}$: The left face ($x_1 = 0$).
    - $\partial_{2,1}$: The back face ($x_2 = 1$).
    - $\partial_{3,0}$: The bottom face ($x_3 = 0$).
- **Degeneracies (σ)**: These squash the 3D cube down into a 2D square by collapsing one of the three dimensions. For example, σ_1^3 drops the x_1 coordinate, pressing the cube flat from left to right.
- **Connections (γ)**: These fold the 3D cube down to a 2D square. Instead of dropping a coordinate entirely, they glue two adjacent faces together (like folding the cube along an internal diagonal plane).

(BONUS: If you want to visualize the underlying ordered structure (poset) for $n = 3$), it looks like a diamond graph:

```
        (1, 1, 1)
       /    |    \
(1, 1, 0)  (1, 0, 1)  (0, 1, 1)
   \    \   |   /    /
    \    \  |  /    /
```

```
(1, 0, 0)   (0, 1, 0)   (0, 0, 1)
    \           |           /
       \        |        /
          (0, 0, 0)
```

(Where the lines represent the monotone increase from 0 *to* 1 *in coordinates).*